Detector-Plane Coherence Transition: A Measurement-Plane Validation of Contextual Wave Imprints via QMCTB-02

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Abstract

We present a quantitative study of interference visibility under detector-plane phase noise using the Quantum Measurement Test Bench (QMCTB-02). Within the Imprint Hypothesis, particles are treated as primary excitations of quantum fields, while wave-like behavior emerges as a contextual imprint arising from particle–field interactions. The Detector Plane Imaging (DPI) framework localizes measurement outcomes at the detector plane.

A double-slit system with Fresnel propagation and detector-level noise modeling is simulated with Gaussian phase noise σ. The system exhibits a sharp transition in visibility at $\sigma_c \approx 2.38$ rad ($R^2 \approx 0.9986$), contrasting with standard continuous decay models.

We interpret this as evidence that interference visibility behaves as a detector-plane transfer function, exhibiting nonlinear threshold behavior governed by correlation fidelity. This establishes a system-level perspective on coherence collapse and provides a testable prediction for experimental systems.

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1. Introduction

The quantum measurement problem is traditionally framed in terms of wavefunction collapse, introducing ambiguity in wave-particle duality.

The Imprint Hypothesis provides an alternative framework:

- Quantum fields are the fundamental substrate
- Particles are primary excitations
- Wave-like behavior is contextual (imprint-based)
- Measurement collapses the imprint, not the particle

The Detector Plane Imaging (DPI) framework further identifies the detector plane as the locus where coherence is preserved or lost.

This work provides a simulation-based validation using QMCTB-02, focusing on detector-plane phase noise and interference visibility.

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2. Theoretical Framework

2.1 Imprint Hypothesis

Field → Particle → Interaction → Imprint → Measurement

Wave-like behavior is interpreted as a contextual imprint arising from particle–field interactions.

2.2 Detector Plane Imaging (DPI)

- Measurement is a structured pipeline
- Detector systems determine correlation fidelity
- Observable outcomes are fixed at the detector plane

2.3 Measurement Principle

Interference visibility is interpreted as a detector-plane response function rather than solely an intrinsic system property.

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3. Methodology (QMCTB-02)

3.1 Simulation Overview

The system models:

- 2D Gaussian source field
- Double-slit aperture geometry
- Fresnel propagation to detector plane

Implemented using NumPy, SciPy, and Matplotlib.

3.2 Detector Modeling

- Pixel binning
- Shot noise (Poisson statistics)
- Read noise (Gaussian)

Gaussian phase noise is introduced at the detector plane:

$\varphi(x, y) \sim N(0, \sigma)$

3.3 Visibility Metric

$V = (I_{max} - I_{min}) / (I_{max} + I_{min})$

Statistical averaging is performed across multiple realizations.

4. Results

- $V(\sigma = 0) \approx 0.91$
- $V(\sigma = \pi) \approx 0.28$
- $\Delta V \approx 0.63$

Sigmoid fit:

- $\sigma\_c \approx 2.38 \pm 0.012$ rad
- $R^2 \approx 0.9986$

This defines a detector coherence transfer function $V(\sigma)$.

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5. Comparison with Prior Work

Standard decoherence models predict continuous decay:

$V(\sigma) \approx V_0 \exp(-\sigma^2 / 2)$

Experimental and theoretical studies (Minář et al., Villar & Lombardo, Kincaid et al.) confirm gradual visibility degradation under Gaussian phase noise.

Nonlinear or oscillatory behavior appears in more complex systems (Mariano et al., Neder et al., Mishra et al.), typically under non-Gaussian or structured noise.

However, sharply defined coherence thresholds under Gaussian phase noise are not commonly reported.

The present work demonstrates a distinct nonlinear transition:

$\sigma\_c \approx 2.38$ rad

suggesting that detector-plane modeling introduces threshold-like behavior beyond standard decoherence descriptions.

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6. Interpretation

6.1 Detector-Plane Transfer Function

Interference visibility behaves as a nonlinear system response determined by detector-plane conditions.

6.2 Coherence Collapse Mechanism

The transition at σ_c corresponds to:

- Loss of phase correlation fidelity
- Collapse of the contextual imprint
- Preservation of particle identity

6.3 Structural Observation

The threshold $\sigma_c \approx 2.38$ rad is numerically close to 2.4048 rad (first zero of J_0).

While no direct equivalence is established, this suggests a possible phase-averaging mechanism underlying the observed transition.

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7. Implications

7.1 Quantum Foundations

- Supports particle primacy over wave duality
- Reinterprets wave behavior as contextual
- Localizes collapse at the detector plane

7.2 Engineering Applications

- Defines coherence tolerance limits
- Guides detector design and optimization
- Enables noise-aware system modeling

7.3 Measurement Theory

Interference visibility emerges as a system response governed by measurement architecture.

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8. Engineering Perspective

The identified threshold $\sigma_c \approx 2.38$ rad is not a universal constant, but a system-dependent boundary emerging from detector-plane configuration and noise structure.

However, its existence suggests a broader principle:

Measurement systems may exhibit nonlinear coherence limits, beyond which signal extraction rapidly degrades.

This has implications for interferometric and phase-sensitive systems, where coherence must be preserved within operational bounds defined by system architecture.

9. Conclusion

This work demonstrates that interference visibility can exhibit threshold behavior under detector-plane phase noise.

The identified coherence boundary:

$\sigma_c \approx 2.38$ rad

is reproducible and system-dependent, indicating that measurement systems can impose nonlinear constraints on observable coherence.

Final Statement

Interference visibility emerges as a detector-plane transfer function exhibiting a coherence threshold under phase noise.

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